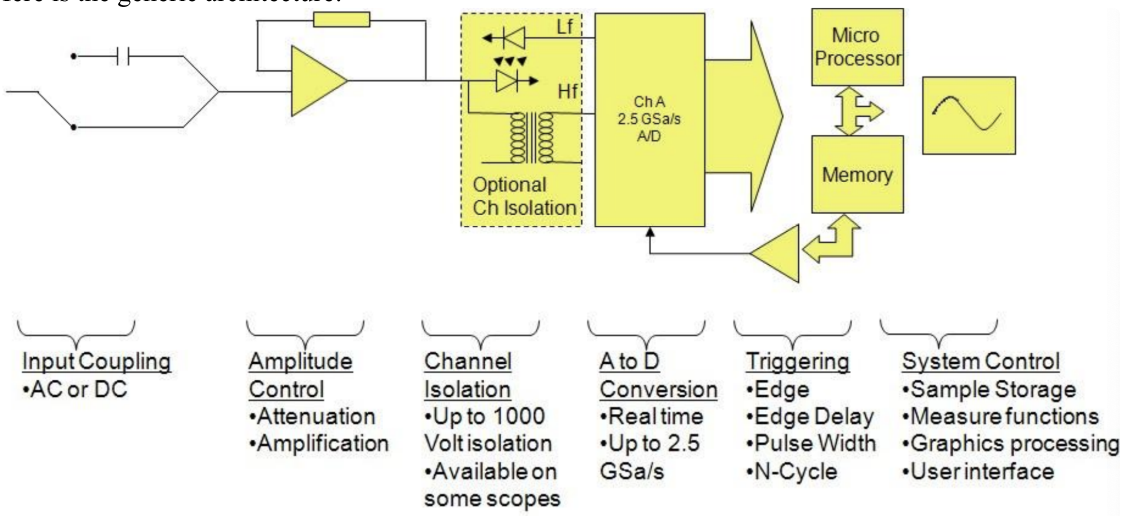
1) Lab Instrumentation

Oscilloscope

The Oscilloscope is a device that graphically displays the electric signals as a function of voltage over time. It is used to measure the wave shape, amplitude, frequency, time between events, noise.

It works by acquiring the signal in an op-amp, samples it, converts the samples to digital, stores and then displays them. Here is the generic architecture:



Architecture

We measure the **performance** with the following parameters:

Definition 1 (Bandwidth)**.**

This is the frequency range of the oscilloscope measured in $[MHz]$. It corresponds with the frequency of the sine wave that is attenuated to 70.7% of the original amplitude.

It is recommended that the oscilloscope has at least 5 times the bandwidth of the desired signal

Definition 2 (Sample Rate)**.**

Since the oscilloscope uses digital samples it must be able to acquire them fast enough. The sample rate is defined as samples/seconds $[S/sec]$. By **Nyquist** it is required that the sample rate is at least twice the highest measured frequency. In practice we need 5 times more

Definition 3 (Record Length)**.**

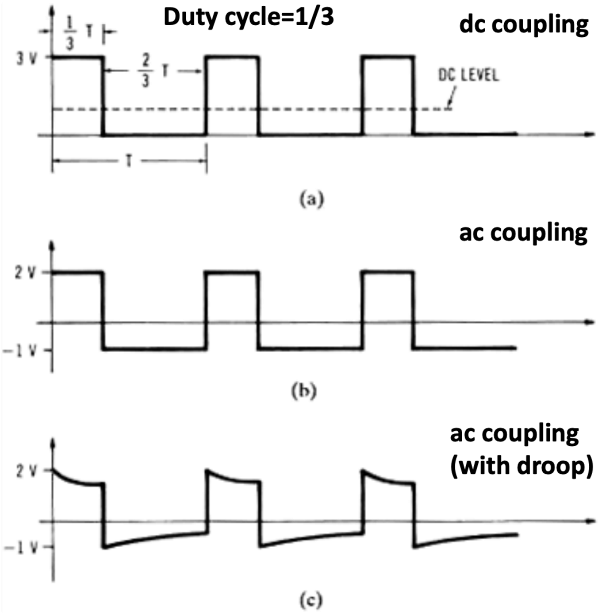
For each acquired waveform the oscilloscope captures a set number of samples.

$$\text{acquired time} = \frac{\text{record length}}{\text{sample rate}}$$

There are multiple controls. Two dilate the scales, one changes the position of the displayed 0 of one signal, one is used for **triggering** and one for **coupling**.

What is it, how does it work?
It is used to horizontally align repetitions of the signal. This allows periodic signals to be displayed correctly without jumping around.

What is coupling, how does it work?
it is divided in AC and DC coupling.
In image (a) we have the standard DC coupling. In image b) AC coupling is applied and therefore the DC level becomes the 0 of the signal, this displays the **original zero-to-peak voltage**. It is not centered since the duty cycle is 1/3. It might also cause **droop** (image (c)) due to the loss of some frequencies.
Moreover it is possible to change the impedance between 1 $[M\Omega]$ and 50 $[\Omega]$. The mega ohm option allows to protect from high input voltages.



Coupling

Function Generator

This simply allows to output a desired waveform

2) Op-Amps

This chapter is not just a recap of op-amps as seen in the electronics course, but it focuses on **performance and limitations** of real life utilization of such devices.
A **ideal** op-amp has infinite gain, wide bandwidth, infinite input resistance, zero output resistance
A **real** op-amp has

	Ideal	**Real**
Gain	∞	finite, $10^5 \div 10^9$
Input resistance	∞	finite, $10^6 \div 10^{12}[\Omega]$
Output resistance	0	non zero $100 \div 1000$ $[\Omega]$
Bandwdth	wide	limited
Slew rate	∞	$0.1 \div 10$ $[V/\mu s]$

Moreover a **real** opamp has also errors given from dc sources (offset voltage, bias current), output voltage and current limits We will see these soon.

Recap

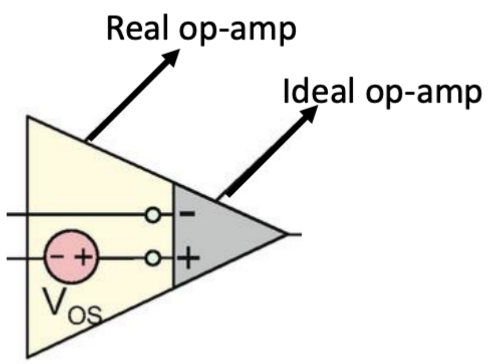
Todo

Input-Offset Voltage

We can measure that with a 0 input **the op-amp rests at some non zero dc value** that is given by

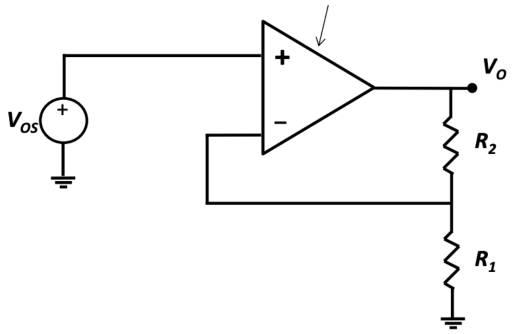
$$V_{OS} = \frac{V_0}{A}$$

Usually we have $v_0 = A(v_{id} + V_{OS})$. Moreover the sign is unknown and there is only an upper bound



Real-Vs ideal

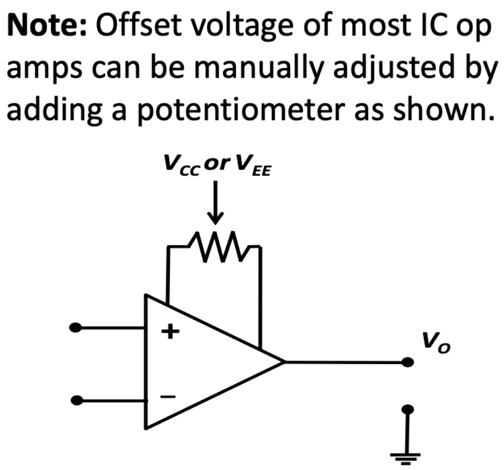
Exercise.



Circuit

Consider the following circuit with $R_1 = 1.2, R_2 = 99\ [k\Omega]$ *and* $|V_{OS}| \leq 3\ [mV]$. *Moreover we assume an ideal op-amp but with non-zero offset voltage.*
Now we have a non inverting op-amp:

$$|V_O| < \left(1 + \frac{R_2}{R_1}\right)|V_{OS}| = 0.25\ [V] \rightarrow -0.25\ [V] < V_O < 0.25\ [V]$$



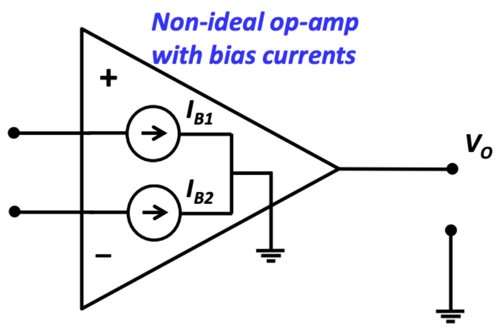
Note

Input-Bias and Offset Currents

As seen in the image a real op-amp has two bias currents, whose values determine the offset current

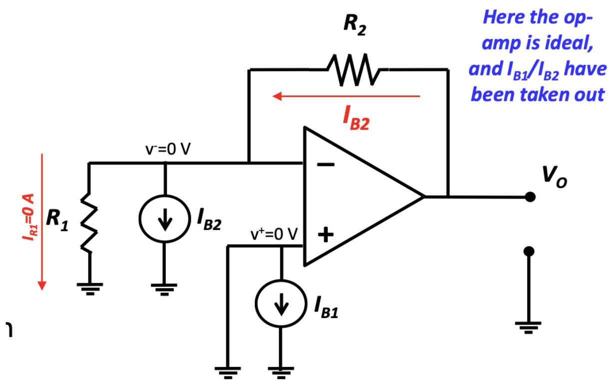
$$I_{OS} = I_{B1} - I_{B2}$$

As before the sign is unknown and there is only an upper bound.



Real op-amp

we can redraw any circuit with a real op amp as a circuit with an ideal op-amp in the following way:



Real to Ideal

This is an inverting configuration where $V_O = I_{B2}R_2$. Where I_{B2} and R_2 can be read in the datasheet as the bias current and input resistance and therefore the value of V_O can be calculated.

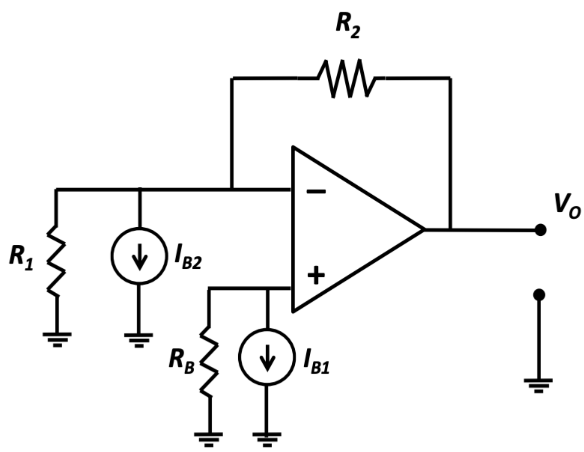
The offset current is 5 to 10 times smaller than the individual bias currents. The dc output voltage error can be reduced by using bias compensation.

We can resort to **bias current compensation** by adding a resistor R_B in parallel with I_{B1} to not short the voltage generator. By superimposition we have that

$$V_{O1} = -I_{B1}R_B\left(1 + \frac{R_2}{R_1}\right)$$
$$V_{O2} = I_{B2}R_2$$
$$\rightarrow V_O = V_{O1} + V_{O2}$$

If we choose

$$R_B = \frac{R_1R_2}{R_1 + R_2}$$



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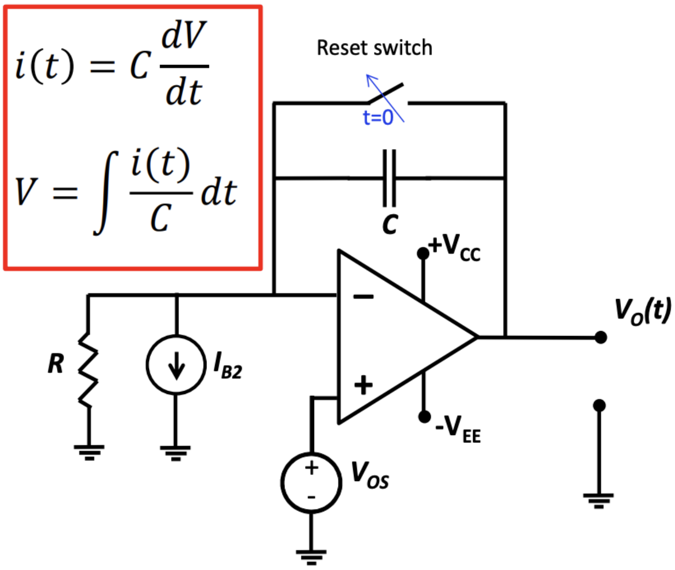
we have that

$$V_O = (I_{B2} - I_{B1})R_2 = -I_{OS}R_2$$

Real Integrator

Now we will study the real integrator circuit at $t = 0$ with the switch just opened.



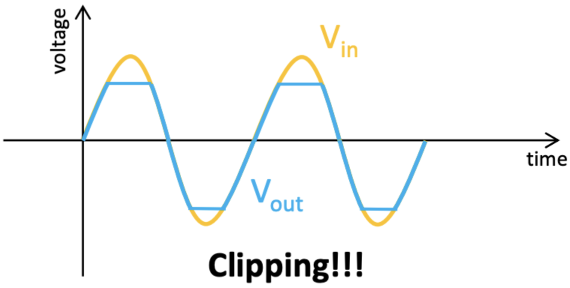
Circuit and Formulas

(As seen before with the switch closed $V_O = V_{OS}$)
Now by solving the circuit we get:

$$v_O = V_{OS} + \frac{V_{OS}}{RC}t + \frac{I_{B2}}{C}t$$

Output Voltage and Current Limits

The **voltage** is limited not directly by the supplied voltage, but it is a few volts shorter.



Voltage Swing

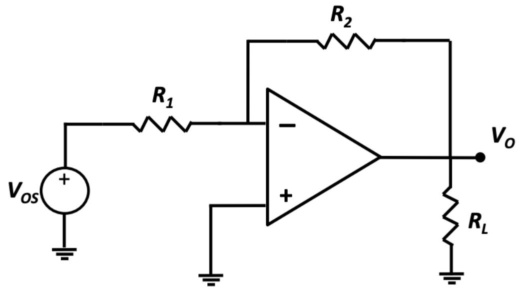
The **current** is limied by additional circuitry to limit power dissipation and/or accidental shorts

Definition 4 (Current Limit)**.**

The current limit is specified as the minimum load resistance that the amplifier can drive with a said voltage swing

$$i_o < \frac{V_S}{R_L}$$

Exercise.



Circuit

Consider the following non-inverting op-amp with $A_v = 20\ [dB]$, $R_L \geq 5\ [k\Omega]$, $v_o \leq 10\ [V]$ and max output current $i_O \leq 2.5\ [mA]$
First of all notice that

$$R_2//R_L \rightarrow R_{EQ} = R_2//R_L \geq \frac{v_O}{i_O} = \frac{10\ [V]}{2.5\ [mA]} = 4\ [k\Omega]$$

By choosing $R_L = 5\ [k\Omega] \rightarrow R_2 \geq 20\ [k\Omega]$ and since $A_v = \frac{R_2}{R_1} = 10$ we can choose $R_1 = 10, R_2 = 100\ [k\Omega]$ to provide an output resistance of $R_{EQ} = 10\ [k\Omega] > 4\ [k\Omega]$. Finally the maximum current will be:

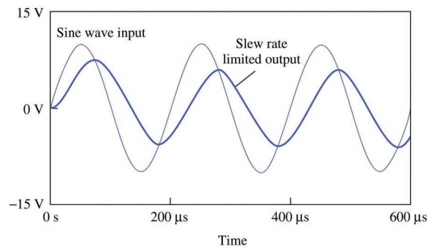
$$i_O \leq \frac{v_O}{R_2} + \frac{v_O}{R_1} = 2.1\ [mA] \leq 2.5\ [mA]$$

While for an inverting amplifier we can use

$$i_O = i_F + i_L = \frac{v_O}{R_L//(R_1 + R_2)}$$

Slew Rate and Limited Bandwidth

The **slew rate** is the maximum rate of change of the op-amp. This will limit the maximum amplitude displayed



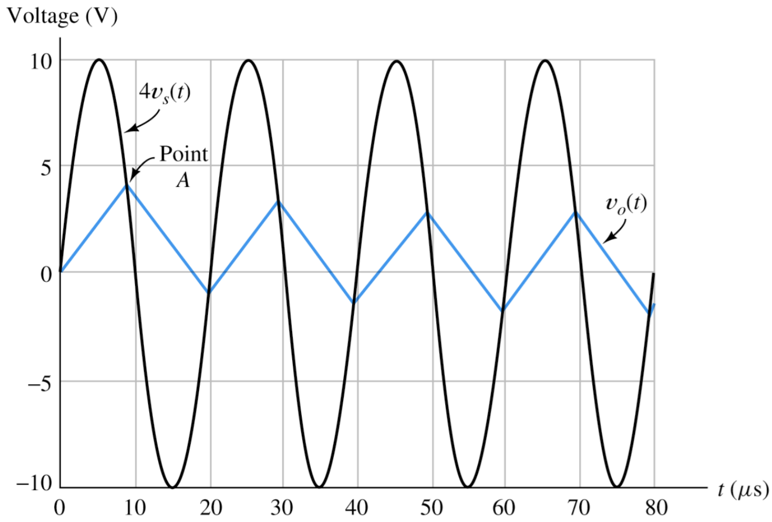
Slew Rate

Fo rno signal distortion we have the following condition:

$$v_O = V_M \sin(\omega t) \rightarrow \frac{dv_0}{dt}|_{\max} = V_M\omega \cos(\omega t)|_{\max} = V_M\omega$$
$$V_M\omega < SR \text{ or } V_M < \frac{SR}{\omega}$$

The full power bandwidth is the highest frequancy that can be displayed

$$f_M = \frac{SR}{2\pi V_{FS}}$$



Example

In this example the output is a triangular wave since we exceed the slew rate of the real op-amp. The black line is the ideal output

Design

If the resistances are too slow we might need an exaggerated amount of voltage, while if they are too big the noise can couple the output. We can do the following:

- limit the gain to a much lower one with respect to te open loop gain ($\approx 10^3$)
- small external resistors with respect to input resistance (input $\approx 1 \div 100$ $[M\Omega]$)
- big resistors with respect to the output resistance (output $\approx 10 - 1000$ $[\Omega]$)
  Then the resistors should be around $1 \div 100$ $[k\Omega]$

Lab Tests

Optional tho

3) Signal Generators

These op-amps based circuits will generate elementary waveforms that can be used in many applications such as clocks, test measurements, information transmission. We will use non-linear oscillators

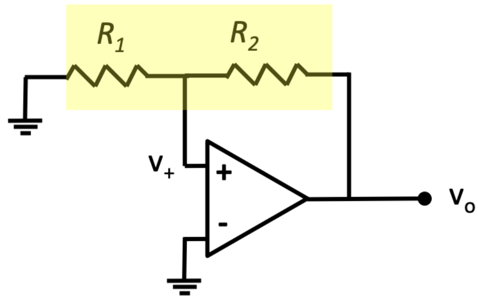
Bistable Circuit (Schmitt Trigger)

A bistable circuit has two stable states that can be changed only when appropriately triggered.

Here we define the parameter β of the voltage divider

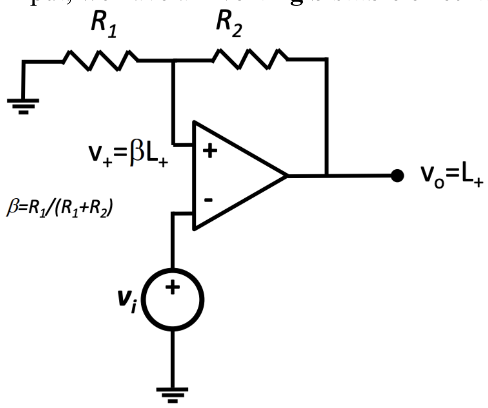
$$\beta = \frac{R_1}{R_1 + R_2}$$

The circuit works by using the **positive feedback**. Suppose we start from $v_+ = 0$ and have a small perturbation that increases the voltage. This signal will now be amplified by the large gain of the op-amp. The voltage divider will feed β to v_+ that will be then again amplified until we reach the stable condition $v_+ = \beta L_+$.



Bistable Circuit

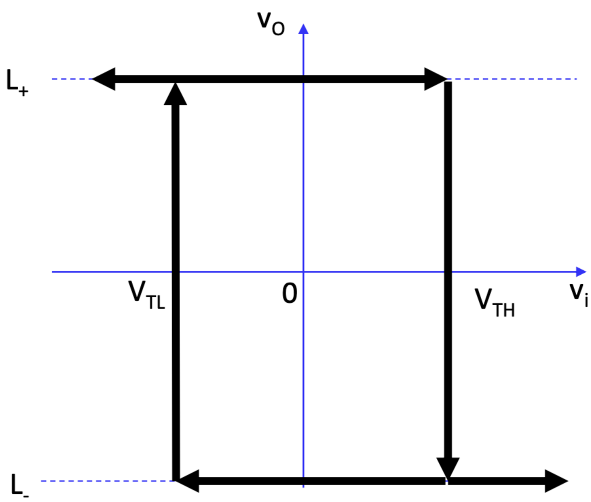
Now add a generator at the negative input, we have a **inverting bistable circuit**:



Inverting Bistable Circuit

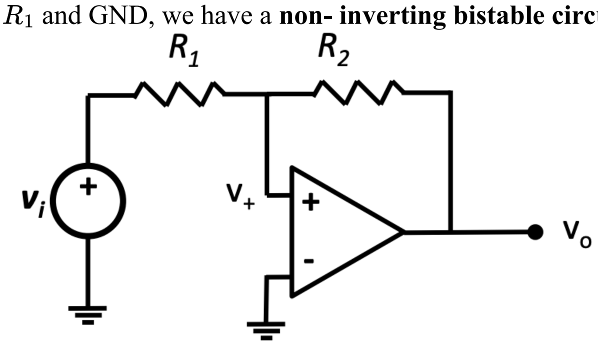
We start with $v_O = L_+$ and thus also $v_+ = \beta L_+$. As we increase v_i from 0 to $v_i = v_+ = \beta L_+ = V_{TH}$ nothing happens. As we increase it further a net **negative voltage appears at the input of the terminals** and v_O goes negative.

To go to the other state we must decrease v_i until below $v_i = v_+ = \beta L_-) = V_{TL}$. In this case a net positive appears and the output spikes to a high value.



> **Remark.**
>
> Once we reach one of the two states we remain with the same output level $\forall v_i$ that do not cross the threshold of the other state. **We have a state memory**

Now add a generator between R_1 and GND, we have a **non- inverting bistable circuit**:



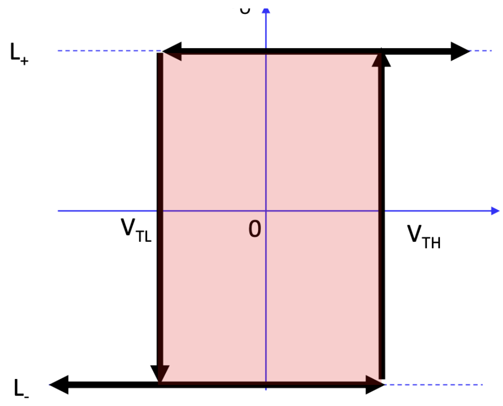
Non Inverting Bistable Circuit

By superimposition we have:

$$v_- = v_i \frac{R_2}{R_1 + R_2} + v_O \frac{R_2}{R_1 + R_2}$$

and thus

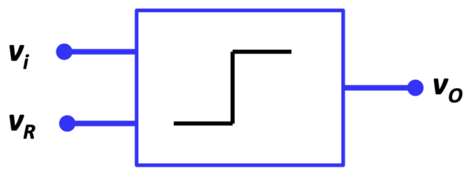
$$V_{TH} = -L_+ \frac{R_1}{R_2}$$
$$V_{TL} = -L_- \frac{R_1}{R_2}$$



Transfer Characteristic

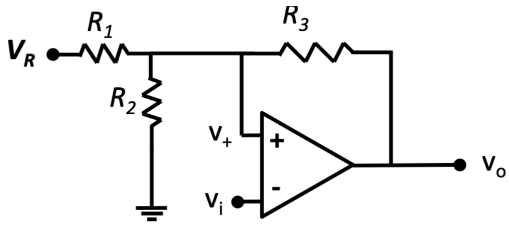
Comparator

The comparator is an analog circuit that detects if a signal is higher or lower than a threshold value. To avoid multiple zeroes due to noise we use **hysteresis, that is, the usage of two thresholds**.



Comparator

This component is built using a **inverting bistable circuit with offset**



Offset Inverting Bistable Circuit

Again, by superimposition, and by defining $V_R' = V_R \frac{R_2//R_3}{R_1 + R_2//R_3}$ and $R' = \frac{R_1//R_2}{R_3 + R_1//R_2}$ we obtain:

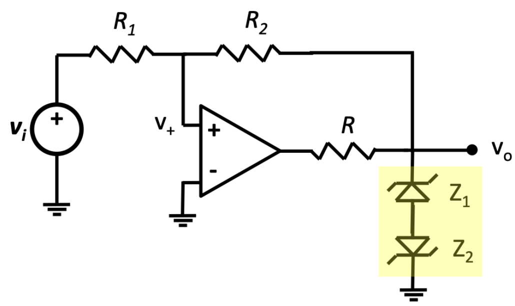
$$v_+ = V_R' + v_O R'$$
$$V_{TH} = V_R' + L_+ R'$$
$$V_{TL} = V_R' + L_- R'$$

Better Accuracy

By adding a **limiter circuit** we can better approximate L_+ and L_-

One approach is given by two back-to-back facing zener diodes and a resistance (that allows zener to work $I_z > I_{z,min}$).
In this configuration we have that

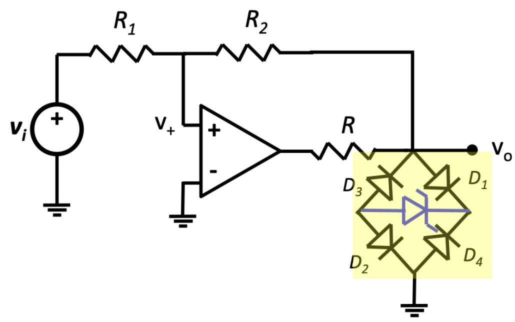
$$L_+ = V_{Z1} + V_D$$
$$L_- = -(V_{Z2} + V_D)$$



Zener Back-to Back

A better approach is the use of a **full-wave bridge rectifier** and a resistance (as before)
In this case we have:

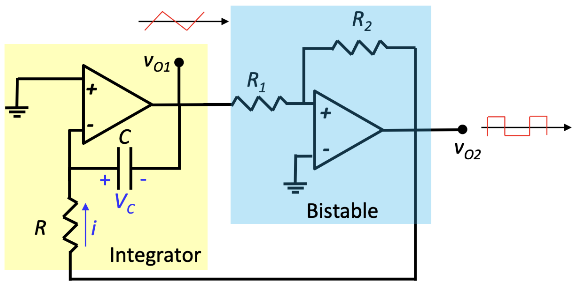
$$L_+ = V_Z + V_{D1} + V_{D2}$$
$$L_- = -(V_Z + V_{D3} + V_{D4})$$



FW BR

Triangular Waveform Generator

By concatenating an integrator and a non inverting bistable circuit it is possible to generate a triangular waveform. The bistable circuit generates square waves as an input to the integrator. By integrating we get a square wave.



Triangular Waveform Generator

Let's analyze this circuit
Start with $v_{O2} = L_+$, on R we have constant $i = L_+/R$. This will cause the integrator to linearly decrease with slope $-L_+/RC$.
Proof:

$$i = C\frac{dV_C}{dt} = \frac{L_+}{R} \rightarrow V_{O1} = V_{O1}(0) - \int \frac{L_+}{RC}dt = V_{O1} - \frac{L_+}{RC}t$$

This continues until $v_{O1} = V_{TL} \rightarrow v_{O2} = L_-$. Now the current on the capacitor will reverse and the output will start to linearly increase until V_{TH}.
Due to the symmetry we have that $L_+ = -L_-$ and thus

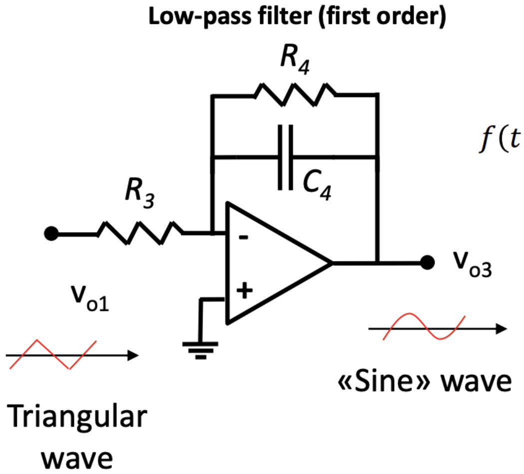
$$T = T_1 + T_2 = 2RC\frac{V_{TH} - V_{TL}}{-L_-}$$

by analyzing the expression of the integrator the times are found by setting $\frac{V_{TH} - V_{TL}}{T_1} = \frac{L_+}{RC}$

numerical example

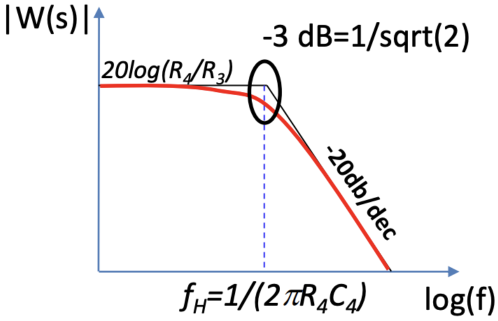
Sine Wave Generator

Here we use a square wave generator (bistable circuit) and a low-pass filter (op-amp)



Low Pass

A simple low pass filter might not be enough since the third harmonic is only attenuated by $\approx 30\%$ as seen in the image.



Bode Plot

Astable Multivibrator

- but it's optional tho

4) Voltage Converters

We will base this chapter on zener regulators:

recap zener regulator

First let's start with a bit of **terminology**
We define the **ability to maintain a constant voltage output in changes in the load as load regulation**. Commercially available PSU have $\approx 1\%$

$$\%\text{Load regulation} = \frac{V_{\text{min load}} - V_{\text{max load}}}{V_{\text{non load}}} \cdot 100$$

The **ability to maintain a constant output voltage with change sin the input as line regulation**

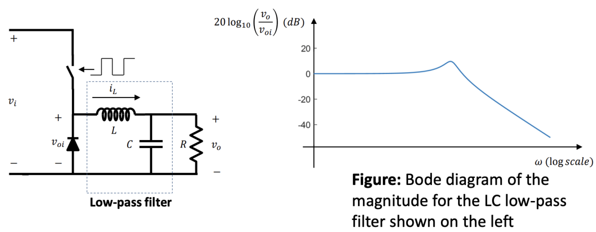
$$\%\text{Line regulation} = \frac{\Delta V_O}{\Delta V_i} \cdot 100$$

DC/DC Converter

A DC/DC Switching Mode Power Converter (SMPC) changes the output by opening and closing a switch. In this way we have a square wave with a duty cycle $\delta = t_{ON}/T$. By integrating we find the average output voltage . By changing duty cycle we can change the avg. output value.

$$V_O = \frac{1}{T}\int_0^{t_{ON}} V_i d\tau + \int_{t_{ON}}^{T} 0 \cdot d\tau = \frac{t_{ON}}{T} V_i = \delta V_i$$

By adding a LC low pass filter we can better aim for a **constant output value** as it limits the harmonic content of the output waveform



Bode And Circuit

The diode solves the problem of the stored inductive energy. The **operating condition** of this circuit is that $I_L > 0$ always. Let's analyze the circuit in the **steady state operation**:

- During t_{ON} the diode is off and the inductor has constant voltage $v_L = V_i - V_O$ and thus the current in the switch is the same one of the inductor that is linearly increasing:

$$v_L = L\frac{di_L}{dt} \rightarrow i_L(t) = i_L(0) + \frac{1}{L}\int v_L(\tau)d\tau = I_{L,min} + \frac{V_i - V_O}{L}t$$

  and thus the variation of the current is

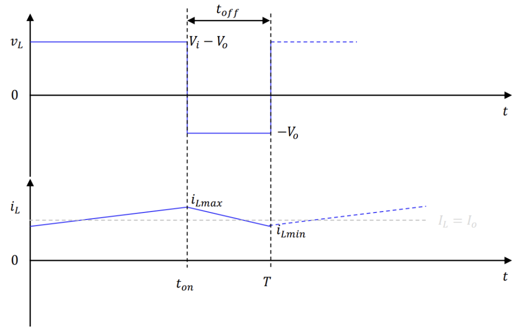
$$\Delta I_{ON} = \frac{V_i - V_O}{L}\delta T = \frac{V_i - V_O}{Lf_S}\delta$$

- During t_{OFF} the current continues flowing through the inductor but the diode turns on. and therefore $v_L = -V_O$. The current decreases as

$$i_L(t) = i_L(0) + \frac{1}{L}\int_0^t v_L(\tau)d\tau = I_{L,max} - \frac{V_O}{L}t \rightarrow I_{L,off} = \frac{V_O}{L}t_{off}$$

It is clear how the current has always the same maximum/minimum $\Delta I_{L,on} = \Delta I_{L,off}$. And the max/min can be found

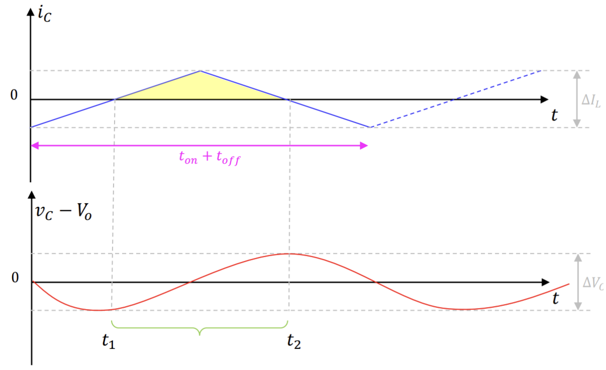
$$i_{L,max} = V_O\left(\frac{1}{R} + \frac{1-\delta}{2Lf}\right)$$
$$i_{L,min} = V_O\left(\frac{1}{R} + \frac{1+\delta}{2Lf}\right)$$



Diagram

Based on this

Now let's study the **voltage ripple:**
The avg current on the capacitor is 0: $I_C = \int_0^T i(c)dt \rightarrow I_C = I_L - I_O = 0 \rightarrow I_L = I_O$



Graph

Consider the interval t_i, t_2. On the capacitor, the voltage ripple is

$$\Delta V_C = \frac{1}{C}\int i_C dt = \frac{1}{C}\int (i_L - I_O)dt = \frac{1}{c}\frac{1}{2}\left(I_{C,max}\frac{t_{on}+t_{off}}{2}\right) = \frac{1}{2C}\frac{\Delta I_L}{2}\frac{T}{2} = \frac{\Delta I_L}{8fC}$$

Final notes:

- In continuous conduction mode (CCM) we always have $I_{L,min} > 0$
- The conversion ratio $V_O = \delta V_i$ does not depend on load and current
- In steady state the avg voltage on the inductor is zero and so is the avg current on the capacitor
- The avg input power is the same as the avg output power (perfect theoretical efficiency)
- The output current is discontinuous and with high harmonic content

Properties of the Filter

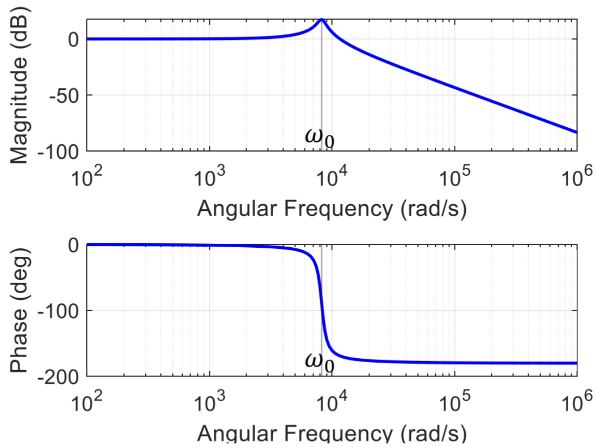
We can evaluate the gain of the filter

$$G_f = \frac{V_2}{V_1} = \frac{\frac{R}{1+sRC}}{\frac{R}{1+sRC} + sL} = \frac{R}{R + sL + s^2LRC} = \frac{1}{s^2LC + s\frac{L}{R} + 1}$$

The poles are

$$s_{1,2} = -\frac{a_1}{2a_2}\left(1 \mp \sqrt{1 - \frac{4a_2}{a_1^2}}\right)$$

with $a_1 = \frac{L}{R}$ and $a_2 = LC$
These gives as a result two complex conjugate poles. That is a underdamped response with cutoff frequency $\omega_0 = \frac{1}{\sqrt{LC}}$
By putting some values we obtain

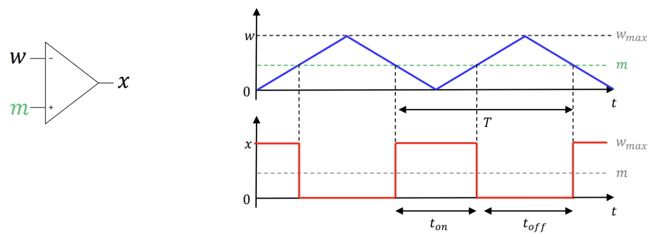


Bode Plots

Control of Output Voltage

We use a comparator to compare a triangular wave with a constant signal w. Then the output value is a square wave with the following duty cycle that depends on the mean of the tri wave:

$$\delta = \frac{t_{on}}{T} = \frac{m}{W_{max}}$$

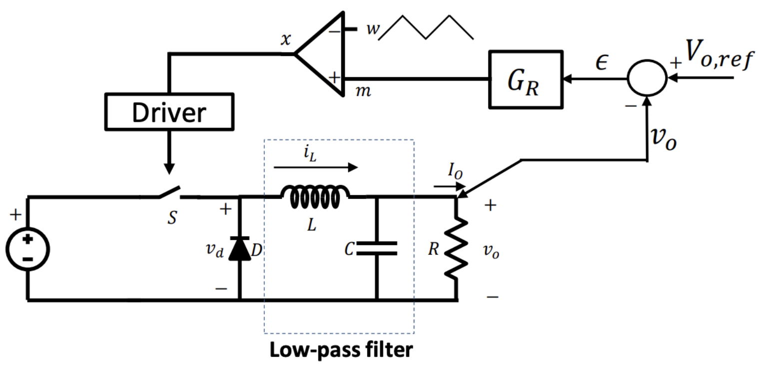


Comparator

This will be used to control the switch of the converter. The working hypothesis is that **m varies at a lower frequency f_m compared to the switching frequency f**

Feedback Loop

The image shows the full feedback loop that will be analyzed, the comparator and LC circuit are already known.



Feedback Loop

1. we analyze v_O with a reference value and we obtain the error ϵ
2. this error is fed to an amplifier G_R whose properties are to assure the stability of the system
3. the output of the amp controls m and thus controls the switch

The transfer function between the duty cycle and v_O is

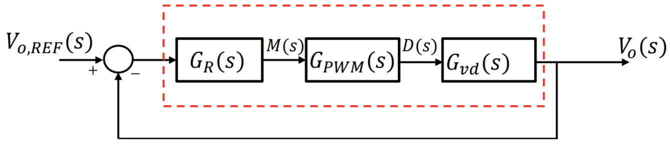
$$G_{vd} = \frac{V_O(s)}{D(s)} = \frac{V_i D G_f}{D} = V_i G_f(s)$$

where $G_f = \frac{1}{1+\frac{s}{Q\omega_0}+\frac{s^2}{\omega_0^2}}$ as discussed with the low pass filter.

And the comparator

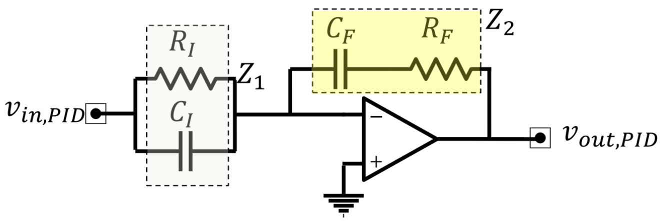
$$G_{PWM} = \frac{D(s)}{M(s)} = \frac{1}{W_{\max}}$$

The full circuit can be modeled as:



Full Circuit

We must be able to ensure that the system is stable and thus G_R **needs to be able to compensate a second order system**. We use a PID controller.



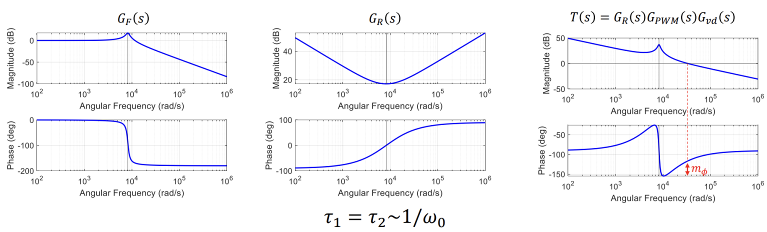
PID

Where $Z_1 = R_I/(1 + sC_I R_I)$ and $Z_2 = 1 + sR_F C_F / sC_F$. from here we have the classic PID expression

$$G_R = -\frac{R_F}{R_I}\frac{(1 + sR_F C_F)(1 + sC_I R_I)}{sC_F R_F} = k\frac{(1 + s\tau_1)(1 + s\tau_2)}{s\tau_1}$$

The full loop is then:

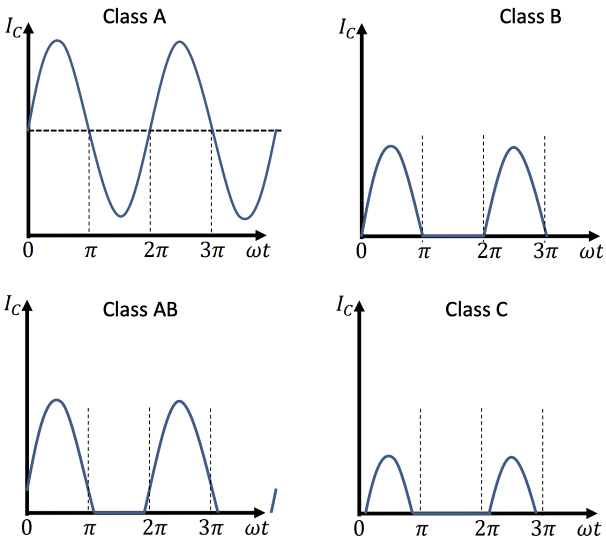
$$T(s) = G_R G_{PWM} G_{vd} = k\frac{(1 + s\tau_1)(1 + s\tau_2)}{s\tau_1}\frac{V_i}{W_{\max}}\frac{1}{s^2 LC + s\dfrac{L}{R} + 1}$$



Final Bode Plots

5) Output Stages

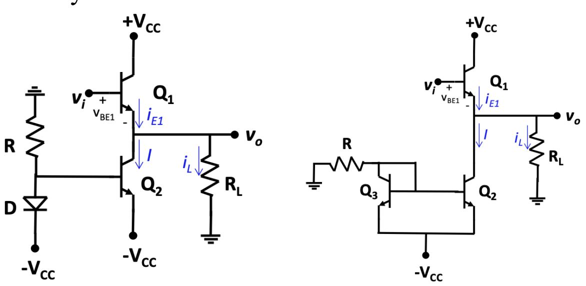
BJT and MOSFET based amplifiers only work with small signals. Real world applications also need to be able to amplify bigger signals (audio amplifiers, RF transmitters). Output stages are large amplifiers. We divide them in 4 classes: A, B, AB, C. The classification is given according to the collector current waveform and the resulting signal



Classes Output

Class A

Class A amplifiers are characterized by their **low internal resistance**.



Diode and Current Mirror based Class A Amplifier

We use the **emitter-follower** approach. The name comes from the fact that $v_{BE1} \approx 0.7$ as long as it is in FAR and thus the variation in **input voltage is the same in the output.**
The transfer function is

$$v_O = v_i - v_{BE1}$$

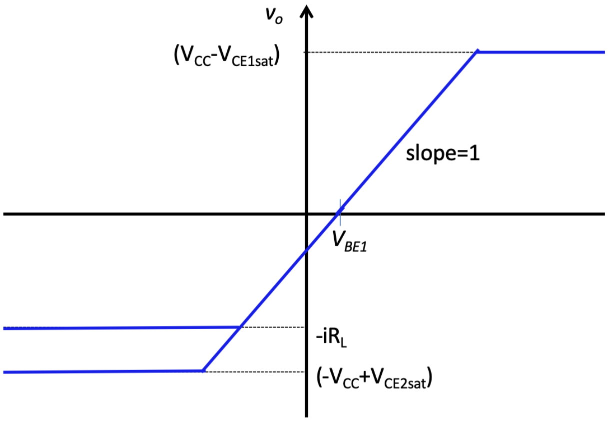
The maximum value is given by reaching the saturation region:

$$v_{O,\max} = V_{CC} - V_{CE1,sat}$$

And the minimum is reached by either the turn off of Q_1 or by the saturation of Q_2

$$v_{O,\min} = -I_{C2}R_L \;\text{ or }\; v_{O,\min} = -V_{CC} + V_{CE2,sat}$$

If we neglect the dependance of v_{BE1} on i_E we obtain the following transfer characteristic



Transfer Characteristic

It is clear that the minimum is reached if $I_E = |-V_{CC} + V_{CE,sat}|/R_L < I$

The current of Q_1 is biased by the one generated by Q_2: $I_{E1} = I_{C2} + I_L$ where $|I_L| < I_{C2}$ to ensure that Q_1 doesn't turn off.

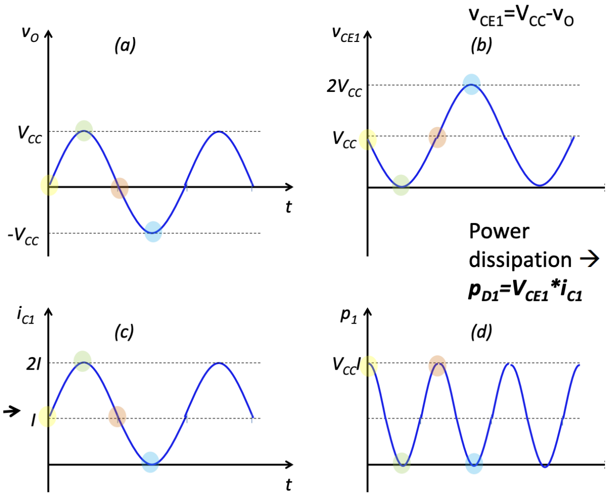
For the diode circuit we have

$$I_{C2} = I_S(e^{V_{BE2}/V_T} - 1)$$

The current mirror allows for a more granular choice in I_{C2} by changing R. But with higher power dissipation. $P_{Q3} = V_{BE3}I$ and $P_{Q2} = (v_O + V_{CC})I$. Where

$$I = \frac{V_{CC} - V_{BE3}}{R}$$

By choosing $I = V_{CC}/R_L$, $V_{CE,sat}$ can be neglected and the output varies between $\pm V_{CC}$



Plots

- The max power is dissipated when $V_o = 0$ since $P_{Q1} = V_{CC}I$
- If we have $R_L = \infty$ then $I_{C1} = I$ and we dissipate $P_{Q1} = 2V_{CC}I$ when $v_O = -V_{CC}$
- Moreover if we have $R_L = 0$ we have a short circuit and thus an infinite(very high) current on the load and thus we break Q_1

On Q_2 we have the maximum dissipation when $V_O = V_{CC}$, the the current is equal to I and the voltage is $2V_{CC}$ then we have $P_{Q2} = 2V_{CC}I$ and it has a minimum at 0.

Definition 5 (Power Conversion Efficiency)**.**

The **power conversion efficiency** is defined as:

$$\eta = \frac{\text{load power}(P_L)}{\text{supply power}(P_S)}$$

In the diode representation $P_L = \frac{V_{OP}^2}{2R_L}$. Since I_2 is constant the power dissipated by Q_2 is $V_{CC}I$ and so is the one on $Q_1 \implies P_S = 2V_{CC}I$
Then the efficiency is

$$\eta = \frac{V_{OP}^2}{2R_L \cdot 2V_{CC}I} = \frac{1}{4}\frac{V_{OP}^2}{IR_LV_{CC}}$$

And is maximum at 0.25 when $V_{Op} = V_{CC} = IR_L$

Pros	**Cons**
Simple	Requires much power (max utilized is 25%)
Low distortion	limited voltage swing

Definition 6 (Total Harmonic Distortion (THD))**.**

THD is the measure of quality of an amp by describing the purity of the output sine wave.

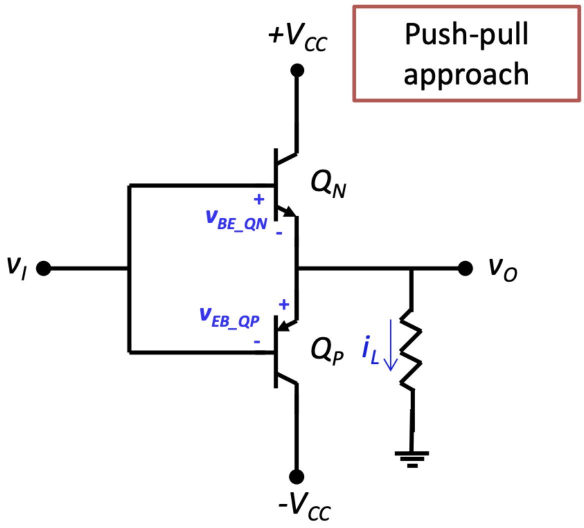
$$THD = \frac{\sqrt{\sum_{i\geq 2} V_i}}{V_1}$$

Class B

A class B amplifier is biased so that it is in linear for 180° of the input cycle, and in cut-off for the remaining 180.
We will see that it has **better efficency** than class A but has more distortion.

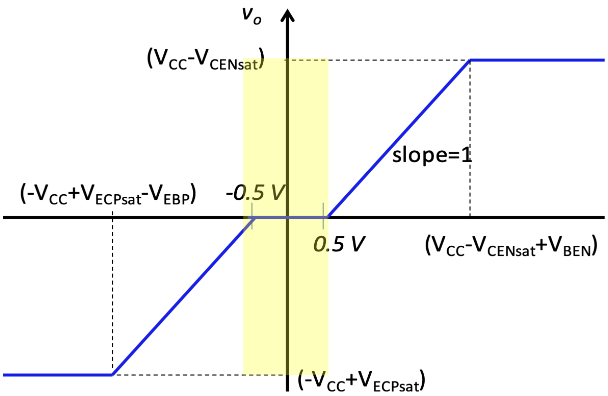
Definition 7 (Push-Pull)**.**

We use push-pull as a term to indicate two resistors that are used on alternating half-cycles



Psuh Pull

With $v_i = 0$ both transistors are off, otherwise $v_i > 0.5$ Q_N behaves as emitter follower ($v_o = v_i - v_{BE,N}$) and at $v_i < 0.5$ the other transistor is the emitter-follower $(v_o = v_i + v_{EB,P})$. The distortion appears at signals $\in [-0.5, 0.5]V$ since the output is null, as also seen by the transfer characteristic

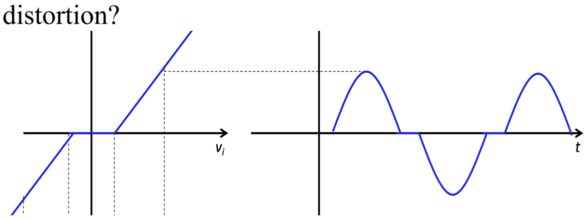


transfer Characteristic

It can be demonstrated that the maximum efficiency is

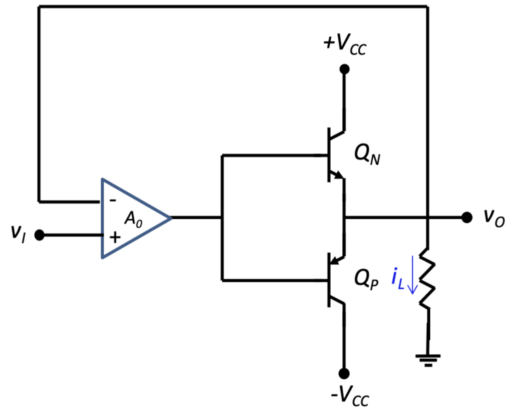
$$\eta = \frac{\pi V_{CC}}{4 V_{CC}} \approx 78.5\%$$

How can we reduce crossover distortion?



Distortion of Pure Sine Wave

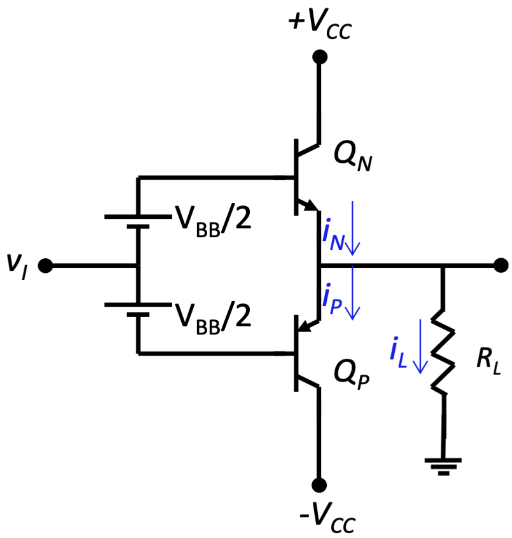
With a high gain opamp and negative feedback we can reduce the bounds to $\pm 0.5/A_0$ but this introduces slew rate limitations and then the alternation is noticeable at high frequencies.



Circuit

Class AB

The distortion can be eliminated by biasing the output transistors at a small non zero current.



Circuit

For $v_o = 0$ we have that $v_{BE} = V_{BB}/2$ and thus (matched devices) $i_N = i_P = I_Q = I_S e^{V_{BB}/2V_T}$
When v_i increases we have that the output becomes positive $v_o = v_i + \frac{V_{BB}}{2} - v_{BE,N}$. Therefore $i_N = i_P + i_L$. since i_L increases **The product** $I_Q^2 = I_N I_P$ **remains constant**. Bo solving the system of the current equations we obtain:$i_N^2 - i_L i_N - I_Q^2 = 0$. this means that:

- Positive output voltages: Q_N supplies the load current and Q_P will decrease in current
- Negative output voltages: the opposite

We will have a small quiescent value

6) Arduino

Let's first analyze the ports:

Ports

Power and Auxiliary I/O

- 5V: This port is the Arduino power and can be used to supply other external devices
- 3.3V: This is also for power and draws a max of 50 mA
- V_{in}: This supplies the same voltage that the arduino is supplied at

Analog Pins

These pins have 10bits of precision from ground to 5 V in steps of 4.9 mV. To have a better accuracy in the arduino due we can go up to 12 bits or use an external ADC with higher resolution

Digital Pins

- Serial: these pins (0 RX, 1 TX) are used for comunication with TTL devices
- PWM: These pins can provide Pulse Width Modulated signals with 8 bit precision
- SPI: These pins (10 SS, 11 MOSI, 13 SCK) are used to provide Serial Peripheral Interface (SPI) communications
- External Interrupts: These pins can trigger interrupts based on the signal they receive
- LED: This pin turns on the on-board led when it is turned on

Since the board is at 500Hz (Uno) the 0% duty cycle is with peaks at 2ms apart

Other

These ports are the RESET, AREF (external reference), TWI (SDA and SCL pin for TWI communication)

On the DUE we can deliver $3 \div 15$ mA and accept $6 \div 9$ mA

Programming vs Native USB Port

- The programming port is reccomended for programming the arduino. It sends a 1200bps hard reset signal
- The native usb port sends a soft erease that can sometimes fail
  They use a bootloader that is a piece of firmware that allows to program over usb directly

Data Types

We program in C++ and since we are on a relatively low performant chip we must ensure to use the right data types to optimize memory usage and performance in calculations.

				Range		
Type	Sign	Bytes	Bits	Min	Max	Other Info.
char	signed	1	8	-128	127	ASCII
char	unsigned	1	8	0	255	ASCII
byte		1	8	0	255	
int (Uno +)	signed	2	16	-32768	32767	Uno model +others.
short		2	16	-32768	32767	
int (Uno +)	unsigned	2	16	0	65535	Uno model +others.
word		2	16	0	65535	Same as unsigned int.
int (Due)	signed	4	32	-2147483648	2147483647	Due model only.
long	signed	4	32	-2147483648	2147483647	Append with 'L'.
int (Due)	unsigned	4	32	0	4294967295	Due model only.
long	unsigned	4	32	0	4294967295	
float		4	32	-3.4028235E+38	3.4028235E+38	6-7 dec digits of precision.
double (Uno +)		4	32	-3.4028235E+38	3.4028235E+38	Same as float.
double (Due)		8	64	(small)	(BIG)	Double precision float.

Data Types Table

Asynchronous Serial Protocol

We can send data with different timing with respect to the main clock. To do so both devices must know the target (baud) rate

Interrupts

Some ports allow to cause a sudden break in the main code and to execute a specific Interrput Service Routine (ISR). This function has some limits as it must return void and can't run for too long, therefore millis() and delay() are disabled. Only delayMicroseconds() can be called but should be avoided. A ISR must be as quick as possible and global variables can be updated to then execute a more complex function during the loop(). It is also useful to classify these variables as **volatile**.

Port Registers

> **Todo**

Serial Peripheral Interface (SPI)

This is a synchronous data protocol used to communicate with one or more peripheral devices. This uses 3 (+1 chip select) lines called

- SDI/MISO: Master In Slave Out, this sends data to the master
- SDO/MOSI: Master Out Slave In, this sends data to the slave
- SCK: Serial ClocK, clock pulse that synchronizes data
- SS: Slave Select, when this goes low the slave will listen for SPI clock and data

This is a single master communication protocol, that is that one device controls multiple other that can communicate only with the master. It selects the slave with the SS, activates the CLK and generates information on MOSi while sampling on MISO. We must then code the arduino with the correct ports for Chip Select (CS), Data Command (DC) and ReSeT (RST)

The bits for the RGB display are 16 bits ordered as 5R6G5B

I^2C Protocol

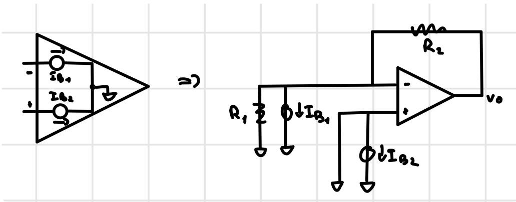
This is again a synchronous protocol. The data is clocked with a clock signal (SCL) that controls when data is changed and when it is read. The clock can vary. It sends two signals:

- SDA: Serial DAta, the data goes here
- SCL: Serial CLock, it is generated by the master and controls when data is sent/read. It can be forced low to block data when a device is oversaturated

7) Microcontrollers

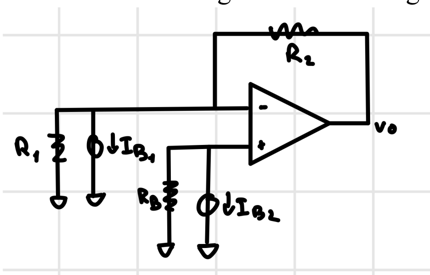
8) Q&A

- **What is the bias current of an operational amplifier? Discuss the impact of bias current on the output voltage of an inverting amplifier. How can the effect be compensated?**
  Real amplifiers have a non zero current flowing through the inputs, these currents are called bias currents and their difference is called offset current. To see how this currents impact the output we can study an inverting configuration with only the bias current as inputs. We will see that with no input voltage, the output will be non zero.



Disegno

v_+ and R_1 are short circuited, and thus we have that $v_O = I_{B1}R_2 \neq 0$.
To reduce this error we can add a resistor in series to change the VSC voltage.



Disegno

Now by superimposition we have:

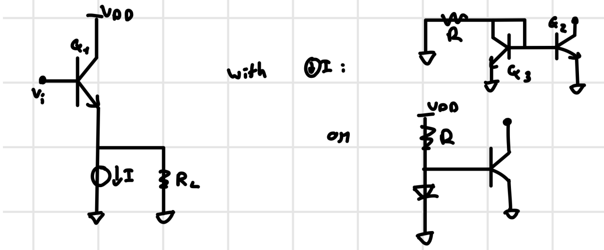
$$v_O|_{I_{B2=0}} = I_{B1}R_2$$
$$v_O|_{I_{B1=0}} = -I_{B2}R_B(1+\frac{R_2}{R_1})$$

with $R_B = R_1//R_2$ we can find that

$$v_O = (I_{B1} - I_{B2})R_2 = I_{OS}R_2$$

which is typically 5 to 10 times smaller than the single bias current.

- **Describe the operation of a class A amplifier, by showing one or more schematic circuits and the transfer characteristic. What is the maximum theoretical efficiency? Discuss with the help of formulas.**
  A class A amplifier operates using the emitter-follower configuration where the output voltage directly follows the input voltage with a slight difference given by the BJT's $V_{BE} \approx 0.7\ [V] \implies v_O = v_i - 0.7$. They are characterized by a low internal resistance and low amplification noise but require a lot of power.



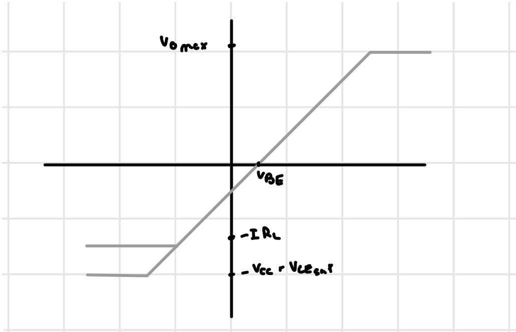
Circuit

  The circuit works until $Q1$ is in FAR and thus we have the following bounds (sat, turn off):

$$v_{o,\max} = V_{CC} - V_{CE,sat} \quad v_{O,\min} = -IR_L$$

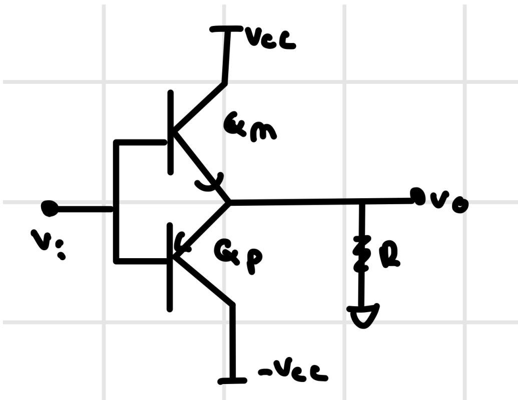
another minimum is obtained by saturating Q_2. $v_{O,\min} = -V_{CC} + V_{CE2,sat}$

Now we can draw the transfer characteristic:



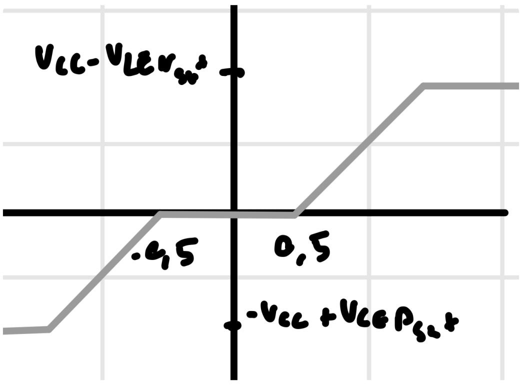
TC

- **Draw the type of amplifier where crossover distortion happens, explain what crossover distortion is, and describe one way to reduce it, by showing the related schematics.**
  This is a type B amplifier:



Disegno

This uses the push-pull principle where only one of the two transistors is always on. The crossover distortion happens in the region hwere both are off, that is $\approx v_i \in [-0.5, 0.5]$. And the upper/lower bounds are given by the saturations of the two transistors $v_{O,\max} = V_{CC} - V_{CEN,sat} \quad v_{O,\min} = -V_{CC} + V_{CEP,sat}$. The transfer characteristic is:



TC

The theoretical maximum is of $\eta = \pi/4$.
We can reduce distortion with

- a high gain negative feedback opamp (bad slew rate)
- a class AB amplifier, that is:
  Adding a $\pm V_{DD}/2$ generator at the transistor base. This completely solves the problem:
- **What is the purpose of the current mirror circuit, when it is used and what is its working principle? explain with the help of schematic.**

Risposta amp di tipo A. Moreover the current mirror acts as a current source. It is possible to better decide the value of the generated current by changing the resistance. The drawback is that it uses mire power since we must drive not only Q2 but also Q3 which fortunately has a constant disspated power of $P_{Q3} = V_{BE3}I$

- **Describe how the communication between two I2C devices works, by summarizing the related signals**
  The Inter-Integrated Circuits protocol is a serial synchronous protocol. It is a 2 BUS protocol since it sends 2 signals: SDA, Serial DAta and SCL, Serial CLock.

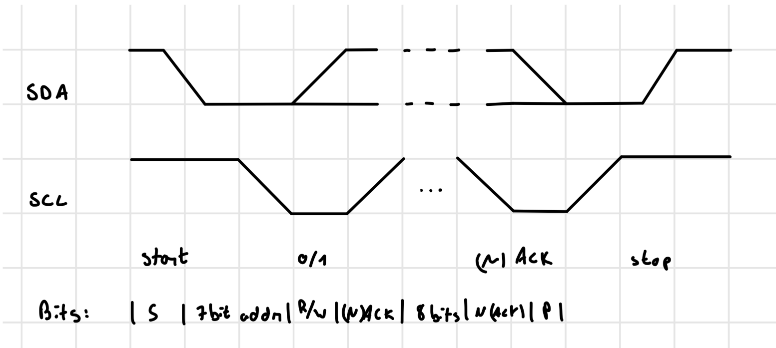
SDA carries the information bits and is able to do so bidirectionally. The SCL is the clock signal generated by the master.

SDA and SCL are both open-drain lines, meaning that they can be pulled low by the devices and are pulled up by the resistors. There are 2 particular conditions

Start condition: SCL is high and SDA is pulled from high to low
End condition: SCL is high and SDA is pulled low to high

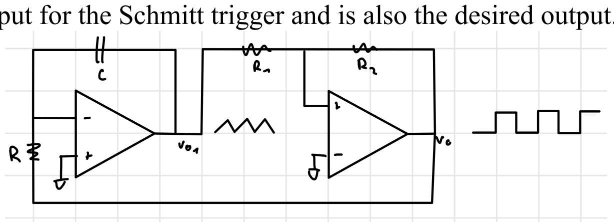
The first communication contains the 7-bit address of the slave to talk to followed by a W/R bit. At the ninth bit we have an ACK that is generated by the slave by pulling SDA low. The other Data is senti in 8 bits with no R/W condition.

The data is sampled when SCL is high and the value is changed when SCL is low, otherwise, if when SCL is high SDA changes we can have the start/stop condition.



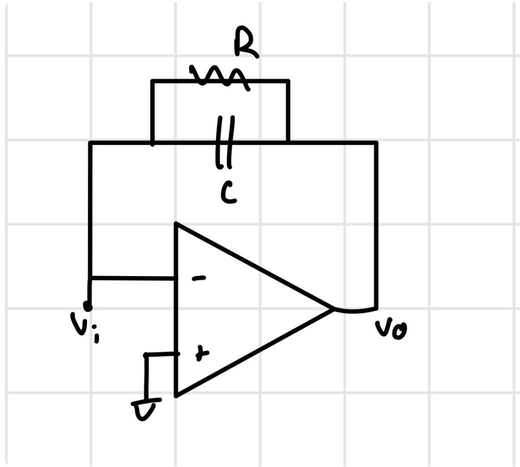
Diagram

- **Explain how we can make a sine wave generator by using a non-linear oscillator that we have studied. Show the related schematics and discuss the operation of the circuit**
  It is easy to see that, by applying a square or sine wave to the input of a Schmitt trigger gives a square wave as an output. By exploiting this we can build a triangular wave generator: we just need to give the output square wave to a integrator that will integrate it into a triangular wave with slope $\pm L_{\pm}/RC$ depending on if it is rising or decreasing. If we have symmetric output values and input thresholds the duty cycle is of 50%.
  This wave is then used as input for the Schmitt trigger and is also the desired output.

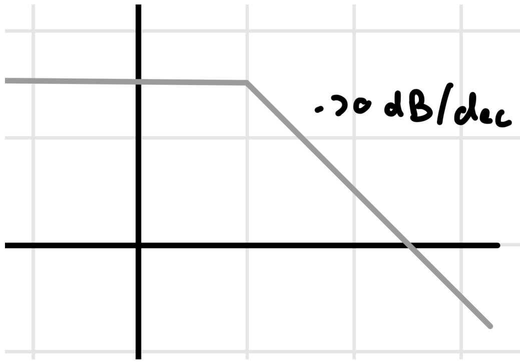


Circuit

  Finally, by recalling the Fourier Series of a triangular wave it is possible to obtain a sine wave by passing the triangular wave through a low pass filter. Unfortunately this approach might need more than one filters since a simple low pass doesn't perfectly attenuate the higher order harmonics. As seen in this example:



Low Pass Circuit



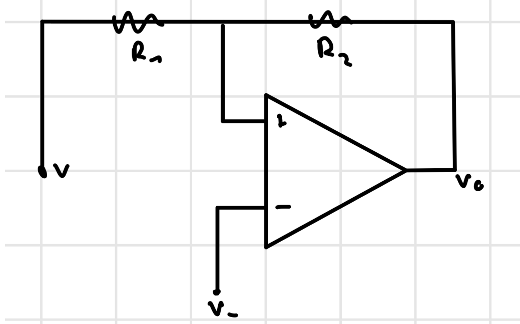
Low Pass Transfer Function

Here we get an attenuation of the third harmonic of only around 30%.

- **What are the differences between the delay and millis functions and why should we use one instead of the other?**
- **Explain the SPI communication protocol, by showing the related signals. What happens to the wiring when multiple peripherals are present?**
  The Serial Peripheral Interface is a synchronous data protocol to communicate with one or more peripherals. there are 3 lines common to all device. Each slave has also a chip select while the master can have one or more chip select lines. The lines/signals are:
- SCK: clock
- MISO/SDI: incoming data
- MOSI/SDO: outgoing data
- CS/SS: chip select

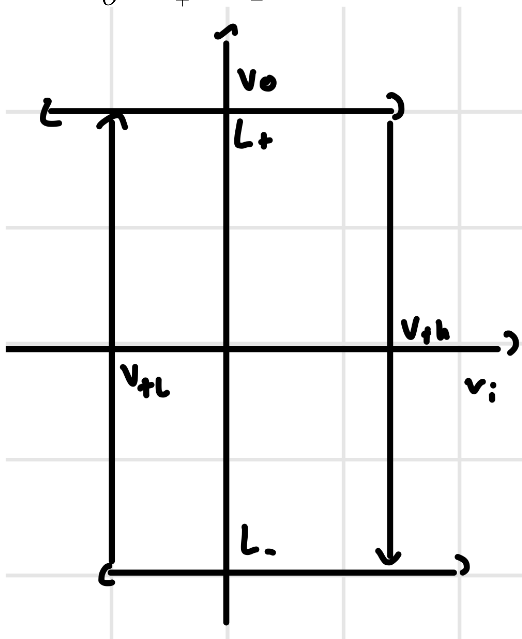
The master selects a slave by pulling down the CS line and sends and receives data simultaneously.

- **Explain what a bistable circuit is. Draw the transfer characteristic and describe the operation.**
  At the heart of the non-linear oscillator is a op-amp that with a positive feedback works as a Schmitt trigger.



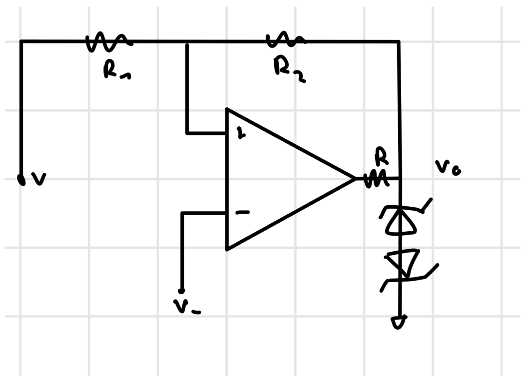
Circuit

Here the two resistor act as a voltage divider with $\beta = R_1/R_1 + R_2$. Moreover we have an inverting configuration if the input is applied at the node $v-$, otherwise if it is applied at the node v we have a non inverting configuration useful for building a comparator.
The name bi-stable comes from the fact that we have 2 stable states. The circuit has memory, this means that once a stable output value is obtained it remains constant at the output and changes only once a threshold value is crossed.
The working principle is based on positive feedback, this means that the output of the amplifier is again re-amplified until a threshold output value $v_O = L_+$ or L_-.



Inverting Transfer function

To have a better precision of $L_\pm$ it is possible to put a two zener diodes back to back or a diode full bridge rectifier.

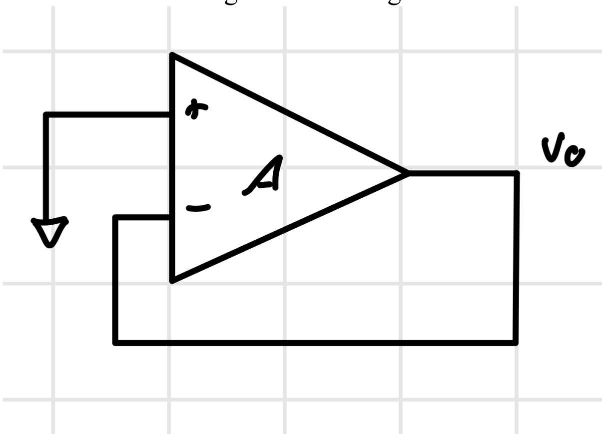


Back To Back Zener Diodes

Here it is clear to see that with symmetric zener diodes $v_O = \pm(V_Z + V_D)$. This works with a resistance that allows $I_O > I_{Z,\min}$.

- **Explain the voltage offset, how to measure it and how to compensate for it.**
  In an ideal opamp when there is no input signal the output is also 0. With real operational amplifiers this isn't true as it is possible to measure a small non zero voltage even with zero input. This is given by $V_{OS} = V_0/A$. Once an input is applied this voltage still influences the output: $v_O = A(v_{id} + V_{OS})$.

To measure this voltage it is possible to use a voltage buffer configuration but with zero input.



Voltage Buffer

To reduce this offset it is possible to use an external potentiometer that has to be adjusted based on the specific parameters of the used op-amp.

- **How does a voltage regulator based on a zener diode work?**
- **What is an Asynchronous Serial protocol?**
  This protocol comes from the words: "Serial", that is sending information in series and "Asynchronous", that is the devices have their own clock but must decide on a bad rate to use before sending data. The Serial() on arduino is the implementation of the serial async protocol UART. During the course this protocol was used with the USB-to-Serial chip to display a log from the arduino on the computer via Arduino IDE, it could also be used to send data from the computer to an arduino.
- **for Amplifier type A, when do we need to use short circuit protection? why?**
- **What is the purpose of the inductor in a dc-dc buck converter?**